

Human Review: Sharp Covering-Index Decay for Schatten Classes

Original automatic proof:	<code>solution_packet (4).pdf</code>
Initial review date:	19 June 2026
Revised review date:	19 July 2026
Status:	Revised manual verification
Classification:	Partial result (downgraded from full result)
Verdict:	Theorem 1 is correct, subject to one minor wording correction. Proposition 1 appears valid. The conclusion of Theorem 2 is recoverable for $1 < p \leq 2$ only after replacing its lower-bound proof; the $p > 2$ claim is unproved.

Verification outcome. This review has been **downgraded to a partial result**. The verified contribution is the explicit Schatten upper bound in Theorem 1, the resulting sharp range $1 < p \leq 2$, and the optimal AUS-exponent statement in Proposition 1. The packet does not establish the sharp covering-index decay for $p > 2$.

Scope of the Revised Review

The original human review concentrated on Theorem 1. It correctly checked the new finite-codimensional rank-one tail argument, but did not independently verify the complemented-subspace monotonicity invoked in Theorem 2. The earlier paragraph titled “Relation to the Lower Bound” and the resulting approval of the packet as a full solution are therefore withdrawn. This revised review checks the lower-bound transfer as well as Proposition 1 and records the corrected classification.

1. Theorem 1: Approved

Theorem 1 proves, for every infinite-dimensional Hilbert space H , every $1 \leq p < \infty$, and every $n \in \mathbb{N}$, that

$$\Theta_{S_p(H)}(n) \leq n^{-1/p}.$$

More precisely, it constructs a cover of $B_{S_p(H)}$ by n closed convex sets, each having essential inradius exactly $n^{-1/p}$.

Write

$$H = H_1 \oplus \cdots \oplus H_n$$

as an orthogonal direct sum of infinite-dimensional closed subspaces, let P_k be the corresponding projections, and set

$$A_k = \{T \in B_{S_p(H)} : \|P_k T P_k\|_p \leq n^{-1/p}\}.$$

The covering step is correct. The finite pinching

$$E(T) = \sum_{k=1}^n P_k T P_k$$

is an average of unitary conjugates of T , and is therefore contractive in every Schatten norm. Since $E(T)$ is block diagonal,

$$\|E(T)\|_p^p = \sum_{k=1}^n \|P_k T P_k\|_p^p.$$

Thus, for $T \in B_{S_p(H)}$, at least one diagonal corner has norm at most $n^{-1/p}$, and the sets A_k cover the unit ball.

The computation of the essential inradius of each A_k is also sound. The inclusion

$$n^{-1/p} B_{S_p(H)} \subseteq A_k$$

gives $\varrho(A_k) \geq n^{-1/p}$. For the reverse inequality, the proof fixes a finite-codimensional subspace

$$Y = \bigcap_{j=1}^m \ker \Phi_j$$

and represents the functionals Φ_j by trace-duality operators B_j . After choosing a finite-rank projection $R \leq P_k$, it selects a rank-one operator $W = \theta_{u,v}$ in the infinite-dimensional tail of the k -th block, with

$$\langle B_j u, v \rangle = 0 \quad (j = 1, \dots, m),$$

so that $W \in Y$. The operators $RV R$ and rW , where $V = P_k T P_k$, act on orthogonal summands, hence

$$\|RV R + rW\|_p^p = \|RV R\|_p^p + r^p.$$

Finite-rank compression and the reverse triangle inequality then show that, for every $r > n^{-1/p}$, one can choose $W \in B_{S_p(H)} \cap Y$ with $T + rW \notin A_k$. Therefore $\varrho(A_k) \leq n^{-1/p}$.

Minor wording correction

The limiting expression in the proof should be written as

$$(\|V\|_p^p + r^p)^{1/p} \geq r > n^{-1/p},$$

not necessarily as a quantity strictly greater than r . Equality occurs when $V = 0$. The corrected inequality is fully sufficient and does not affect the proof.

2. The Lower-Bound Transfer in Theorem 2 Is Invalid

The packet identifies the diagonal subspace $D \cong \ell_p$ of $S_p(H)$ and the contractive diagonal conditional expectation onto D . Those facts are correct. The proof then invokes the claimed monotonicity principle

$$\Theta_D(n) \leq \Theta_{S_p(H)}(n)$$

for contractively complemented subspaces. That principle is not valid in general, so the diagonal copy of ℓ_p does not by itself yield the required lower bound for $S_p(H)$.

Error in the cited version-1 lemma

Lemma 3.3 of version 1 of Kania–Mařłany starts with a near-optimal cover

$$B_Y = \bigcup_{k=1}^n B_k$$

of the complemented range Y , and lifts it to

$$A_k = \{x \in B_X : Px \in B_k\}.$$

The argument correctly proves

$$\varrho_Y(B_k) \leq \varrho_X(A_k).$$

It then uses the inequality

$$\Theta_X(n) + \varepsilon \geq \max_k \varrho_X(A_k).$$

This step is unjustified: the lifted sets A_k were not chosen as a near-optimal cover of B_X . By the definition of the infimum, one only has

$$\Theta_X(n) \leq \max_k \varrho_X(A_k),$$

which points in the opposite direction and gives no comparison between $\Theta_Y(n)$ and $\Theta_X(n)$.

The current version 2 of the Kania–Mařłany paper, revised 6 March 2026, no longer contains that monotonicity lemma. Its related Bochner-space conclusion is formulated only as an upper-bound result, and its appendix explicitly leaves the precise sharp covering-index decay for Schatten classes open.

Counterexample to the proposed monotonicity

The monotonicity statement is in fact false. Let

$$X = c_0 \oplus_\infty \ell_2, \quad Y = c_0,$$

where Y is the range of the norm-one coordinate projection. Choose a Hilbert-space cover

$$B_{\ell_2} = \bigcup_{k=1}^n C_k$$

with $\varrho_{\ell_2}(C_k) = n^{-1/2}$, and define

$$A_k = B_{c_0} \times C_k.$$

Then the A_k cover B_X , and

$$\varrho_X(A_k) = \varrho_{\ell_2}(C_k).$$

The lower inequality follows by lifting a finite-codimensional ball in C_k through the subspace $c_0 \oplus V$. For the reverse inequality, intersect any finite-codimensional subspace of X with $\{0\} \oplus \ell_2$ and recover a finite-codimensional ball in C_k . Consequently,

$$\Theta_X(n) \leq n^{-1/2}.$$

On the other hand, Raja records

$$\Theta_{c_0}(n) = 1$$

for all n . Thus, for $n \geq 2$,

$$\Theta_Y(n) = 1 > n^{-1/2} \geq \Theta_X(n),$$

although Y is contractively complemented in X . This disproves the monotonicity principle used in Theorem 2.

3. Corrected Covering-Index Conclusions

The valid upper bound from Theorem 1 combines with the established non-commutative AUS lower bound

$$\Theta_{S_p(H)}(n) \gtrsim n^{-1/r}, \quad r = \min\{p, 2\}.$$

The corrected conclusions are therefore as follows.

For $1 < p \leq 2$,

$$\Theta_{S_p(H)}(n) \asymp n^{-1/p}.$$

Here the non-commutative lower bound has exponent $1/p$, so it matches Theorem 1. At $p = 2$, $S_2(H)$ is itself a Hilbert space, and the exact value is

$$\Theta_{S_2(H)}(n) = n^{-1/2}.$$

For $2 < p < \infty$, the justified bounds are only

$$c_{p,H} n^{-1/2} \leq \Theta_{S_p(H)}(n) \leq n^{-1/p}$$

for some $c_{p,H} > 0$. The exponents do not match. The packet's claimed lower bound of order $n^{-1/p}$ for this range is unsupported, and the sharp decay remains open.

Theorem 1 also remains valid at $p = 1$, but the packet does not supply a verified matching lower bound at that endpoint.

4. Proposition 1: Approved Separately

Proposition 1 claims that the largest AUS power exponent obtainable by an equivalent norm on $S_p(H)$, for $1 < p < \infty$, is

$$\min\{p, 2\}.$$

This argument appears valid and does not depend on the false complemented-subspace monotonicity.

The natural Schatten norm is r -AUS for $r = \min\{p, 2\}$ by the non-commutative Clarkson inequalities, so exponent r is attainable. To exclude larger exponents, one restricts a hypothetical q -AUS renorming to suitable subspaces. For $1 < p \leq 2$, the diagonal copy of ℓ_p rules out $q > p$ by comparison with the known sharp scalar estimate $\Theta_{\ell_p}(n) \asymp n^{-1/p}$. For $p \geq 2$, a row or column subspace is Hilbertian, and the exact Hilbert value $\Theta_H(n) = n^{-1/2}$ rules out $q > 2$.

For cleaner complex-linear notation, one may use the column space

$$\{\theta_{u,e_0} : u \in H\}$$

in place of the row-space notation, depending on the convention for the inner product.

5. Required Revisions to the Solution Packet

1. Change the packet's status from a full result to a **partial result**.

2. Retain Theorem 1, with the minor correction $(\|V\|_p^p + r^p)^{1/p} \geq r > n^{-1/p}$.
3. Replace Theorem 2 by the sharp statement for $1 < p \leq 2$ and the nonmatching bounds for $p > 2$.
4. Remove the claim that contractive complementation transfers the scalar ℓ_p lower bound to $S_p(H)$.
5. Retain Proposition 1 as a separately verified result, with updated theorem numbers and slightly cleaner row/column notation.
6. Update the citations to the current source version: the exact Hilbert calculation is Theorem 2.3 and the scalar L_p conclusion is Corollary 3.3 in version 2.

Verification and Classification

Verification classification: Partial result (downgraded).

The previous full-solution designation is retracted. The durable verified result consists of:

1. the explicit upper bound $\Theta_{S_p(H)}(n) \leq n^{-1/p}$, with a cover whose pieces have exact essential inradius $n^{-1/p}$;
2. the resulting sharp asymptotic decay $\Theta_{S_p(H)}(n) \asymp n^{-1/p}$ for $1 < p \leq 2$;
3. the optimal equivalent AUS power exponent $\min\{p, 2\}$.

The sharp covering-index decay for $2 < p < \infty$ is not verified by this packet. The attempted proof rests on a false general monotonicity claim, so this part must remain open unless a new Schatten-specific lower-bound argument is supplied.

Final Review Outcome

Approve only as a partial result. Theorem 1 is correct after the minor wording amendment, and Proposition 1 appears correct. Theorem 2 is valid in the range $1 < p \leq 2$ after replacing its lower-bound proof by the established non-commutative AUS lower bound. The asserted matching decay for $p > 2$ is not proved and must not be presented as a full solution of the Schatten sharp-decay problem.

References

- [1] T. Kania and N. Mařlany, *Raja's covering index of L_p spaces*, arXiv:2512.13249v1.
- [2] T. Kania and N. Mařlany, *Raja's covering index of L_p spaces*, arXiv:2512.13249v2, revised 6 March 2026.
- [3] M. Raja, *A covering index for Banach spaces*, arXiv:2503.03721.
- [4] *Sharp Covering-Index Decay for Schatten Classes*, automatic solution packet dated 16 June 2026.